

DNA Recombination - Factoid Sheet

Week 4 - Tuesday - Core Lecture 5 - current faculty deck + official Week 4 objectives

Master chain: DSB or DNA movement → choose recombination pathway → repair/rearrange DNA → genetic variation or disease.

Four Types

Type	Rule	Anchor
HR	Long homology; template directed	High-fidelity DSB repair; meiosis
NHEJ	Little/no homology	G1/somatic DSB repair; V(D)J
Transposition	Mobile element changes position	DNA transposons / retrotransposons
Site-specific	Defined sequence sites	Precise rearrangement systems

Homologous Recombination

DSB → resection → 3' strand invasion → Holliday junction → branch migration → resolution. Spo11 creates programmed meiotic DSBs.

Consequences Crossovers, gene conversion, new alleles, accurate DSB repair.	Failure Nondisjunction/aneuploidy; infertility with Spo11 defects; NAHR structural rearrangements.
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Linkage / HLA

Closer loci recombine less often. HLA genes are tightly linked and generally inherited as haplotypes. A full sibling has a classic 25% chance of inheriting the same two parental HLA haplotypes.

HR vs NHEJ

HR	NHEJ
Template dependent	Template independent
High fidelity	Error prone
S/G2; meiosis	Especially G1; somatic cells
Misaligned repeats → NAHR	Wrong ends joined → translocation

NHEJ Clinical Anchors

CML: t(9;22) BCR-ABL. Burkitt lymphoma: t(8;14) MYC. NHEJ defects can impair V(D)J recombination and cause severe immunodeficiency.

DNA Recombination - Fast Recall

Transposons

Element	Mechanism	Key fact
DNA transposon	Cut-and-paste	Transposase; no inherent copy-number increase
Retrotransposon	RNA intermediate + RT; copy-and-paste	Copy number increases; repetitive DNA expands
LINE-1	Retrotransposon	Autonomous
Alu / SINE	Retrotransposon	Non-autonomous; uses LINE-1 machinery

Processed Pseudogene

Cellular mRNA accidentally reverse transcribed and reinserted. Typical clues: **no introns**, lacks original promoter, may have 3' poly(A)-derived tract, may have target-site duplications, often nonfunctional.

Antibody Diversity

RAG1/RAG2 recognize RSS and create DSBs. **NHEJ** rejoins coding ends. Junctional imprecision adds diversity. Cleavage chemistry resembles ancient cut-and-paste transposition.

Repetitive DNA Disease Logic

Repeated homologous sequences can misalign → non-allelic HR → deletion, duplication, inversion, translocation. Deck anchors include PMP22 dosage disease and factor VIII inversion.

Must-Not-Miss Traps

HR needs homology; NHEJ does not.	HR is repair + meiotic exchange, not just crossover.
LINE autonomous; Alu non-autonomous.	Processed pseudogenes lack introns because the intermediate was spliced mRNA.

30-Second Review

HR = homology/high fidelity. NHEJ = direct/error prone. HLA = tight linkage. Repeats = NAHR risk. Retrotransposons use RNA + RT. RAG cleavage + NHEJ joining = V(D)J diversity.

Assigned ScholarRx

Chromosomal Translocations, Deletions & Inversions: <https://exchange.scholarrx.com/brick/chromosomal-translocations-deletions-and-inversions>